

Vector Calculus

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Lecture 1 – 23/01/26

§1 Preliminaries

§1.1 Vector notation

We will in general work in $\mathbb{R}^3$, but sometimes $\mathbb{R}^2$. In fact, most of our results will generalise to $\mathbb{R}^n$.

We will write column vectors to save space when we mean row vectors, and we will use dot notation for differentiation (even though t isn't really time).

If $\mathbf{a}$ and $\mathbf{b}$ are vectors, then the following hold:

$$\mathbf{a}(t) = \frac{d\mathbf{a}}{dt} = \frac{d}{dt}(a_1, a_2, a_3) = (\dot{a}_1, \dot{a}_2, \dot{a}_3) \quad (1)$$

$$\frac{d}{dt}(\mathbf{a} \cdot \mathbf{b}) = \dot{\mathbf{a}} \cdot \mathbf{b} + \mathbf{a} \cdot \dot{\mathbf{b}} \quad (2)$$

$$\frac{d}{dt}(\mathbf{a} \times \mathbf{b}) = \dot{\mathbf{a}} \times \mathbf{b} + \mathbf{a} \times \dot{\mathbf{b}} \quad (3)$$

and these results can be shown using suffix notation. Integration is also componentwise. We have therefore

$$\int_{t_1}^{t_2} \dot{\mathbf{a}}(t) dt = \mathbf{a}(t_2) - \mathbf{a}(t_1)$$

The modulus of a vector will be denoted by the same letter as the vector, but not bolded/underlined. So, for example, $r = |\mathbf{r}|$ (but x means x_1 , not $|\mathbf{x}|$).

Notationally, we might write ∂_i for $\frac{\partial}{\partial x_i}$.

§1.2 The chain rule

Recall that, if f is a function of $\mathbf{x}$ which is itself a function of t , then

Proposition 1.1

$$\begin{aligned}\frac{d}{dt}f(\mathbf{x}(t)) &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \\ &= \frac{\partial f}{\partial x_i} \frac{dx_i}{dt}\end{aligned}$$

If instead $\mathbf{x} = \mathbf{x}(\mathbf{u})$, where $\mathbf{u} = (u_1, u_2, u_3) = (u, v, w)$, then

$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial u}$$

and similarly for $\frac{\partial f}{\partial v}$ and $\frac{\partial f}{\partial w}$, i.e.

$$\frac{\partial f}{\partial u_i} = \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial u_i}$$

§1.3 Taylor's theorem

Recall that for a small vector displacement $\delta\mathbf{x}$,

$$\begin{aligned}f(\mathbf{x} + \delta\mathbf{x}) &= f(x + \delta x, y + \delta y, z + \delta z) \\ &= f(x, y, z) + \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial z} \delta z + \frac{1}{2!} \left(\delta x \frac{\partial}{\partial x} + \delta y \frac{\partial}{\partial y} + \delta z \frac{\partial}{\partial z} \right)^2 f + \dots\end{aligned}$$

or, in suffix notation,

$$f(\mathbf{x} + \delta\mathbf{x}) = f(\mathbf{x}) + \frac{\partial f}{\partial x_i} \delta x_i + O(|\delta\mathbf{x}|^2)$$

§1.4 Infinitesimals

In one dimension, we have from Taylor's theorem

$$\delta f = f(x + \delta x) - f(x) = f'(x) \delta x + O(\delta x^2)$$

Taking $\delta x \rightarrow 0$, δx and δf become dx and df , and we get the exact equality

$$df = f'(x) dx$$

If we want, we can even think of infinitesimals as numbers such that their squares are zero.

Note that infinitesimals have their own product and quotient rules, namely

$$\begin{aligned}d(fg) &= g df + f dg \\ d\left(\frac{f}{g}\right) &= \frac{g df - f dg}{g^2}\end{aligned}$$

We also get

Proposition 1.2

$$df = \frac{\partial f}{\partial x_i} dx_i \quad (1.3)$$

The modulus of an infinitesimal vector can be calculated using, for example

$$|d\mathbf{x}| = |dx| \sqrt{1 + \left(\frac{dy}{dx}\right)^2 + \left(\frac{dz}{dx}\right)^2}$$

If dx and dy are independent infinitesimals then $dx dy$ is an infinitesimal area. But if y depends on x in some way then $dx dy$ vanishes, since it's proportional to dx^2 .

§2 Vector differential operators

Remark. What is an operator? It is a map between vector spaces of functions.

§2.1 Del

The fundamental differential operator is *del*:

$$\nabla = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix}$$

We can also write

$$\nabla = \mathbf{e}_1 \frac{\partial}{\partial x_1} + \mathbf{e}_2 \frac{\partial}{\partial x_2} + \mathbf{e}_3 \frac{\partial}{\partial x_3} = \mathbf{e}_i \frac{\partial}{\partial x_i}$$

where $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is the canonical basis for $\mathbb{R}^3$.

§2.2 Gradient

When *del* is applied to a scalar field $f(\mathbf{x})$, it is known as the *gradient* operator:

$$\nabla f = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix}$$

This can also be denoted $\text{grad} f$.

In suffix notation, we have

$$\begin{aligned} (\nabla f)_i &= \frac{\partial f}{\partial x_i} \\ \iff \nabla f &= \frac{\partial f}{\partial x_i} \mathbf{e}_i \end{aligned}$$

- Example** • If $f(\mathbf{x}) = |\mathbf{x}|^2 = x^2 + y^2 + z^2$, then $\nabla f = (2x, 2y, 2z) = 2\mathbf{x}$. Alternatively, $\frac{\partial}{\partial x_i}(x_j x_j) = 2x_i \delta_{ij} = 2x_i$ (the δ_{ij} comes from $\frac{\partial x_j}{\partial x_i}$).
- $\nabla |\mathbf{x} - \mathbf{x}_0|^2 = 2(\mathbf{x} - \mathbf{x}_0)$. This is a direct consequence of the above and a very simple application of the chain rule.
 - $\nabla(\mathbf{a} \cdot \mathbf{x}) = \nabla(a_1 x + a_2 y + a_3 z) = a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_3 = \mathbf{a}$. Alternatively, $\frac{\partial}{\partial x_i}(a_j x_j) = a_i \delta_{ij} = a_i$.

∇f in 3D is analogous to f' in 1D, and we can evaluate it at points.

Example

If $f(\mathbf{x}) = x + y^2 + z^4$, then $\nabla f(\mathbf{a}) = \nabla f(\mathbf{b})$ if and only if $\mathbf{a} - \mathbf{b}$ is parallel to the x -axis. Observe $\nabla f(\mathbf{x}) = (1, 2y, 4z^3)$, so

$$\begin{aligned} (1, 2a_2, 4a_3^3) &= (1, 2b_2, 4b_3^3) \\ \iff (a_2, a_3) &= (b_2, b_3) \\ \iff \mathbf{a} - \mathbf{b} &= (a_1 - b_1, 0, 0) \end{aligned}$$

We have some new opportunities for notation now – for example, we can rewrite (1.1) as

Proposition 2.1

$$\frac{d}{dt} f(\mathbf{x}(t)) = \dot{\mathbf{x}} \cdot \nabla f$$

and (1.3) as

Proposition 2.2

$$df = \nabla f \cdot d\mathbf{x}$$

Equation (2.2) also gives a way of defining ∇f that is *coordinate-independent*, i.e. it doesn't require us to choose a particular basis at any point.

Example

If $f(x) = |\mathbf{x}|^2$, then

$$\begin{aligned} df &= f(\mathbf{x} + d\mathbf{x}) - f(\mathbf{x}) = \mathbf{x} \cdot \mathbf{x} + 2\mathbf{x} \cdot d\mathbf{x} + d\mathbf{x} \cdot d\mathbf{x} - |\mathbf{x}|^2 \\ &= 2\mathbf{x} \cdot d\mathbf{x} \end{aligned}$$

so it must be the case that $\nabla f = 2\mathbf{x}$.

Finally, we may write $\nabla_{\mathbf{x}}$, $\nabla_{\mathbf{y}}$, etc., to denote differentiation w.r.t. a particular vector variable.

§2.3 Divergence

Definition. The *divergence* of a vector field is given by del dotted with the field:

$$\nabla \cdot \mathbf{F} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = \frac{\partial F_i}{\partial x_i}$$

More formally, we can derive this expression by

$$\begin{aligned} \nabla \cdot \mathbf{F} &= \left(\mathbf{e}_i \frac{\partial}{\partial x_i} \right) \cdot (F_j \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot \frac{\partial}{\partial x_i} (F_j \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot \frac{\partial F_j}{\partial x_i} \mathbf{e}_j \\ &= \frac{\partial F_j}{\partial x_i} \delta_{ij} = \frac{\partial F_i}{\partial x_i} \end{aligned}$$

What is the divergence? It measures the amount by which a field is ‘expanding’ around $\mathbf{x}$. We will get a better idea later on when we derive a coordinate-independent definition.

Example • Take $F(\mathbf{x}) = \mathbf{x}$. Then $\nabla \cdot F = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 3$. Suffix notation also works! DIAGRAM

- Take $F(\mathbf{x}) = (-y, x, 0)$. Then $\nabla \cdot F = -\frac{\partial y}{\partial x} + \frac{\partial x}{\partial y} = 0$. DIAGRAM
- If A is a 3 by 3 matrix, we might want to calculate

$$\nabla \cdot (A\mathbf{x}) = \frac{\partial}{\partial x_i} (A_{ij}x_j) = A_{ij}\delta_{ij} = A_{ii} = \text{Tr } A$$

§2.4 Curl

Definition. The *curl* of a vector field is given by del crossed with the field:

$$\nabla \times \mathbf{F} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \begin{pmatrix} \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \\ \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \\ \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \end{pmatrix}$$

In suffix notation, we have

$$(\nabla \times F)_i = \varepsilon_{ijk} \frac{\partial}{\partial x_j} F_k$$

The curl, notably, is only defined in 3D, since we only have a cross product in 3D, unlike gradient or divergence. It measures, in a certain sense, the *rotation* of the field about a point $\mathbf{x}$.

Example • With $F(\mathbf{x}) = \mathbf{x}$, we have

$$\nabla \times F = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{0}$$

Alternatively, $(\nabla \times F)_i = \varepsilon_{ijk} \frac{\partial}{\partial x_j} x_k = \varepsilon_{ijk} \delta_{jk} = 0$.

- With $F(\mathbf{x}) = (-y, x, 0)$, $\nabla \times F = (0, 0, 2)$.
- If A is a 3 by 3 symmetric matrix,

$$(\nabla \times (A\mathbf{x}))_i = \varepsilon_{ijk} \frac{\partial}{\partial x_j} A_{kl} x_l = \varepsilon_{ijk} A_{kj}$$

But ε_{ijk} is antisymmetric while A_{kj} is symmetric, so the terms cancel in pairs and we get $\nabla \times (A\mathbf{x}) = \mathbf{0}$.